

ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3 \times 10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m²)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D/(4\pi)$ — effective aperture area (m²)
- A_p — physical cross-section area (m²)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)

Key shortcuts: Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{\text{rad}} = 73 \Omega$. 3 dB $\Rightarrow$ linear gain of 2.

Problem 1 — Hertzian Dipole, Power Density at 5 km

8 pts

Setup: A 1-m long dipole excited by a 1 MHz current with amplitude $I_0 = 12$ A. Find the average power density radiated at $R = 5$ km, $\theta = 45^\circ$.

DIPOLE TYPE BY LENGTH

Compute first $\lambda = c/f$, then ratio l/λ		
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
 $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi} \frac{e^{-jkR}}{R} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
 Power density $l/\lambda < 1/50$; far field
 $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\max} \sin^2 \theta$ (W/m²)
 $k = 2\pi/\lambda$ (rad/m); I_0 peak current (A); l length (m); R obs. dist (m); $\eta_0 = 120\pi \approx 377 \Omega$.
 Max direction broadside $\theta_{\max} = \pi/2$
 Total radiated power integrate over 4π
 $P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2$ (W)
 Directivity exact $D_{\text{Hertz}} = 1.5$ (1.76 dB)

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

Half-wave is the default “dipole antenna” unless stated otherwise.

Step 1 — Classify the dipole length. Compute the wavelength to compare with l :

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \text{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ($l/\lambda < 1/50$) means we can use the Hertzian $S(R, \theta)$ formula directly.

Step 2 — Apply the Hertzian power density formula. From HERTZIAN DIPOLE — $S(R, \theta)$:

$$S(R, \theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta$$

Step 3 — Plug in numbers. $\eta_0 = 120\pi$, $k = 2\pi/\lambda = 2\pi/300 \text{ rad/m}$, $I_0 = 12 \text{ A}$, $l = 1 \text{ m}$, $R = 5000 \text{ m}$, $\theta = \pi/4$:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5 \times 10^3)^2} \sin^2(\pi/4)$$

$$\text{Numerator: } (120\pi)(4\pi^2/9 \times 10^4)(144) = (120\pi)(144)(4\pi^2)/(9 \times 10^4)$$

$$= 120 \cdot 144 \cdot 4 \cdot \pi^3 / (9 \times 10^4) = 69120\pi^3 / (9 \times 10^4)$$

$$\text{Denominator: } 32\pi^2 \cdot 2.5 \times 10^7. \text{ And } \sin^2(\pi/4) = 1/2.$$

Working it out cleanly:

$$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2$$

Sanity: a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m². The $1/R^2$ falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

Problem 2 — Beam Parameters from Normalized Pattern

15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \leq \theta \leq \pi$ (θ in radians). Find: (a) half-power beamwidth β , (b) pattern solid angle Ω_p , (c) directivity D .

BEAM PARAMETERS — β , Ω_p , D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *solve $F(\theta_{1/2}) = 0.5$, then $\beta = 2\theta_{1/2}$ for symmetric beams*

Gaussian shortcut: $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$.

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$$

No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.

Most integrals have NO closed form. Use Wolfram / numerical.

Wolfram: `2*pi*Integrate[F[t]*Sin[t],{t,0,pi}]`

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$$

Alt. with two orthogonal HPBW's: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$.

Step 5 — Gain (if asked) $\xi = \text{rad. eff.}, 0 < \xi \leq 1$

$$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

(a) Half-power beamwidth β (5 pts). From BEAM PARAMETERS Step 2, the original cheat sheet equations:

$$F(\theta_{1/2}) = 0.5 \Rightarrow \text{solve for } \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams)}$$

Plug in $F(\theta) = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5)$$

$$\theta_{1/2} = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = 0.186 \text{ rad}$$

That's the half-angle from the peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full half-power beamwidth* is twice that:

$$\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ$$

(b) **Pattern solid angle Ω_p (5 pts).** From BEAM PARAMETERS Step 3:

$$\Omega_p = \int \int_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi$$

F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta$$

By hand (narrow-beam Gaussian shortcut from cheat sheet). The beam is narrow ($\beta \approx 21^\circ$), so use $\sin \theta \approx \theta$ and extend the upper limit to ∞ (the Gaussian dies off fast):

$$\Omega_p \approx 2\pi \int_0^\infty e^{-20\theta^2} \theta \, d\theta$$

Let $u = 20\theta^2 \Rightarrow du = 40\theta \, d\theta$, so $\theta \, d\theta = du/40$:

$$= 2\pi \int_0^\infty e^{-u} \frac{du}{40} = \frac{2\pi}{40} \int_0^\infty e^{-u} \, du = \frac{\pi}{20}$$

$$\boxed{\Omega_p \approx \frac{\pi}{20} \approx 0.157 \text{ sr}}$$

Textbook numerical answer is 0.156 sr; the narrow-beam approximation is within 1%.

(c) **Directivity D (5 pts).** From BEAM PARAMETERS Step 4:

$$D = \frac{4\pi}{\Omega_p}$$

Using the by-hand $\Omega_p \approx \pi/20$:

$$D \approx \frac{4\pi}{\pi/20} = 4 \cdot 20 = 80$$

$$\boxed{D \approx 80}$$

Textbook value using $\Omega_p = 0.156$ exactly: $D = 4\pi/0.156 = 80.55$.

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern

15 pts

Setup: Dipole of length $l = \lambda/4$. Find: (a) directions of maximum radiation, (b) expression for $S_{\max}$, (c) plot of normalized pattern $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$ Power density at R any l

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$ Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$$

Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, *shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}$$

$$D = 1.64, \quad R_{\text{rad}} = 73 \Omega, \quad \theta_{\max} = \pi/2.$$

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

Half-wave is the default “dipole antenna” unless stated otherwise.

General arbitrary-length dipole pattern. From ARBITRARY-LENGTH DIPOLE — $S(\theta)$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Plug in $l = \lambda/4$, so $\pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2 = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Let $S_0 = 15I_0^2/(\pi R^2)$ be the constant prefactor.**(a) Direction of maximum radiation (5 pts).** Maximizing $S(\theta)/S_0$ for $0 \leq \theta \leq \pi$. By calculus or numerical optimization, the max occurs at:

$$\theta_{\max} = \pi/2 = 90^\circ$$

(broadside to the dipole axis, same as half-wave). At $\theta = 0$ or π (along axis), $S = 0$.**(b) $S_{\max}$ (5 pts).** Evaluate $S(\pi/2)$:

$$S_{\max} = S(\pi/2) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cdot 0\right) - \frac{\sqrt{2}}{2}}{\sin(\pi/2)} \right]^2 = \frac{15I_0^2}{\pi R^2} \left(1 - \frac{\sqrt{2}}{2}\right)^2$$

Numerically $(1 - \sqrt{2}/2)^2 = 0.0858$:

$$S_{\max} = 0.0858 \frac{15I_0^2}{\pi R^2}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts):

$$F(\theta) = \frac{1}{0.0858} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π . Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Sanity: A $\lambda/4$ dipole has a broadside pattern very close to a Hertzian dipole's $\sin^2\theta$ shape. As $l \rightarrow 0$ the bracket approaches $\sin\theta$ and $S \propto \sin^2\theta$ (Hertzian). Half-wave $l = \lambda/2$ has slightly narrower beam than the $\lambda/4$ case.

Problem 4 — Effective Area vs. Physical Cross-section

5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find effective area A_e and compare to physical cross-section A_p .

EFFECTIVE AREA A_e

Definition
antenna's RX cross section (m^2)

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section
wire dipole, diameter d

$$A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{rad} \text{ (}\Omega\text{)}$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

Half-wave is the default “dipole antenna” unless stated otherwise.

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

Step 2 — Directivity for half-wave dipole. From COMMON DIPOLE PARAMETERS: $D = 1.64$.

Step 3 — Effective area. From EFFECTIVE AREA A_e :

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2 (1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$A_e = 1.17 \text{ m}^2$$

Step 4 — Physical cross-section. A half-wave dipole has length $l = \lambda/2$. Treat as a rectangle of length l and width d :

$$A_p = l \cdot d = \frac{\lambda}{2} \cdot d = \frac{3}{2} (0.02) = 0.03 \text{ m}^2$$

Step 5 — Ratio.

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2$$

Sanity: The effective area is $39\times$ larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region $\sim \lambda$ around itself.

Problem 5 — Friis Transmission, TV Broadcast

5 pts

Setup: Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{dB}/10}$.3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.Use linear G , not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{dB} = 10 \log_{10} G_{lin} \quad G_{lin} = 10^{G_{dB}/10}$$

	dB	linear	dB	linear
	0	1	10	10
	3	2	13	20
	6	4	20	100
	7	5	30	1000
	-3	0.5	-10	0.1

Memorize: 3 dB $\leftrightarrow \times 2$, 10 dB $\leftrightarrow \times 10$. Add dB = multiply linear.For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

Step 2 — Antenna gains. TX is half-wave dipole, so $G_t = 1.64$ (from COMMON DIPOLE PARAMETERS).

RX gain: 3 dB $\rightarrow$ linear via $G_{lin} = 10^{G_{dB}/10}$:

$$G_r = 10^{3/10} = 2$$

(3 dB always = $\times 2$, this is the cleanest dB conversion to remember.)**Step 3 — Friis formula.** From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Step 4 — Plug in. $R = 30 \text{ km} = 3 \times 10^4 \text{ m}$, $P_t = 10^3 \text{ W}$:

$$P_r = (1.64)(2) \left(\frac{6}{4\pi(3 \times 10^4)} \right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi \cdot 3 \times 10^4} = \frac{6}{3.77 \times 10^5} = 1.59 \times 10^{-5}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 2.53 \times 10^{-10}$$

$$P_r = (1.64)(2)(2.53 \times 10^{-10})(10^3) = (3.28)(2.53 \times 10^{-7})$$

$$P_r = 8.3 \times 10^{-7} \text{ W} = 0.83 \mu\text{W}$$

Sanity: 1 kW transmitted, 0.83 μW received at 30 km. That’s a 90 dB drop, dominated by the $1/R^2$ free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is $-100 \text{ dBm} = 10^{-13} \text{ W}$), so this link closes comfortably.

Note on the solution PDF: the official key wrote “(1.64)(3)” as a typo, but used $G_r = 2$ in the actual arithmetic (since 3 dB $\rightarrow$ linear = 2). The final answer $8.3 \times 10^{-7} \text{ W}$ matches either way.